

5 (a) (i) Define *current*.

.....  
 ..... [1]

(ii) Hence, show that

$$I = nAve$$

where  $I$  is the current,  $n$  is the number of conduction electrons per unit volume,  $A$  is the cross sectional area of the conductor,  $v$  is the average drift velocity and  $e$  is the elementary charge.

[2]

(iii) A potential difference of 0.2 V is applied across a 3.0 m long copper wire with a uniform circular cross sectional diameter of 1.0 mm. The electron density and resistivity of copper can be taken to be  $8.49 \times 10^{28} \text{ m}^{-3}$  and  $1.68 \times 10^{-8} \Omega \text{ m}$  respectively.

1. Determine the resistance of the copper wire.

resistance of the copper wire = .....  $\Omega$  [2]

2. Hence or otherwise, determine the average drift velocity of the electrons.

drift velocity of the electrons = .....  $\text{m s}^{-1}$  [3]

- (iv) The potential difference is now applied across a tapered conductor as shown in Fig 5.1.

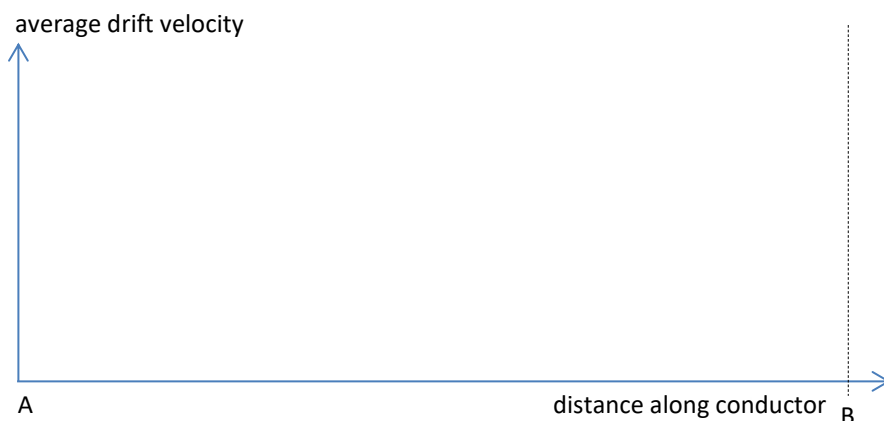


**Fig 5.1**

Sketch on Fig 5.2a and Fig 5.2b, the variation of current and the average drift velocity of the electrons along the conductor.



**Fig 5.2a**

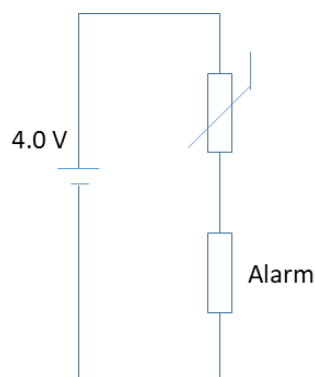


**Fig 5.2b**

[3]

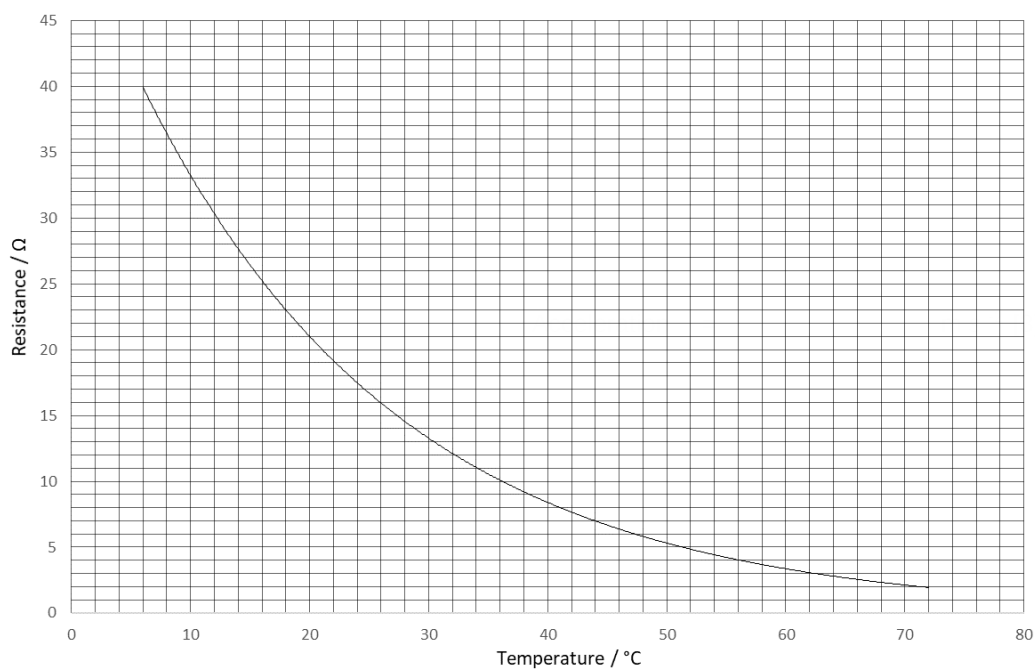
**[Turn over**

- (b) A student constructed a heat detector to be used as a fire alarm by connecting a 4.0 V dry cell in series with a thermistor and an alarm as shown in Fig 5.3. The alarm has a resistance of  $30\ \Omega$  will be triggered when the potential difference across it is more than 3.6 V.



**Fig 5.3**

The variation of temperature with resistance of the thermistor is as shown in Fig 5.4.



**Fig 5.4**

- (i) Determine the minimum temperature that will trigger the alarm.

minimum temperature = ..... °C [2]

- (ii) The student suggests modifying the heat detector by attaching a flashing light in parallel to the alarm.

Explain the change, if any, to the minimum temperature that will trigger the alarm.

.....

.....

.....

..... [2]

- (iii) A smoke detector, rather than a heat detector, is usually used as a fire alarm. Suggest a possible reason why a heat detector is not suitable to be used as a fire alarm.

.....

..... [1]

**[Turn over**